FUNCTION GENERATOR

EC LAB PROJECT

GROUP MEMBERS:

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INTRODUCTION:

An op-amp based function generator is a useful tool in electronics laboratory applications as it is capable of producing waveforms with variable frequency and amplitude. The basic motivation for designing such a function generator using op-amps is its simplicity, low cost, and versatility. Op-amps are widely available and can be easily integrated into circuit designs, making them an ideal choice for generating a wide range of signals for testing and experimentation purposes. Additionally, they offer high accuracy and stability, allowing for control of waveform shape, frequency, and amplitude. Overall, an op-amp based function generator offers a cost-effective and reliable solution for generating various waveforms in a laboratory setting.

THEORY:

Here, three components are used to create the op-amp based generator:

1.Wein-Bridge oscillators

2.Bi-stable multivibrator (Schmitt trigger)

3.Integrator

WEIN BRIDGE OSCILLATOR:

Wein Bridge Oscillators are electronic circuits used to generate sine waves at a frequency determined by the components used in the circuit. The theory behind Wein Bridge Oscillators is based on the principle of positive feedback or regeneration of a signal through a frequency selective network to produce oscillations.

The basic circuit of a Wein Bridge Oscillator consists of an op-amp, a feedback network, and a frequency-selective network. The feedback network usually consists of a resistor and a capacitor in series, and the frequency-selective network consists of two resistors and two capacitors arranged in a bridge configuration.

When the circuit is powered, the op-amp will amplify the signal fed back from the feedback network. At the same time, the bridge network will act as both a frequency-selective filter and a phase shift network.

The circuit is designed in such a way that the feedback signal and the signal from the bridge network are in phase and are fed back to the input of the op-amp. The positive feedback causes the output of the op-amp to oscillate at a frequency determined by the values of the components used in the frequency-selective network.

The oscillation frequency can be adjusted by varying the values of the resistors and capacitors in the bridge network or the feedback network. Wein Bridge Oscillators are commonly used in audio and radio frequency applications where a stable and precise sine wave is required.

BI-STABLE MULTIVIBRATOR:

A bistable multivibrator is a circuit that has two stable states: high and low. It is also known as a flip-flop circuit. Bistable multivibrators are very commonly used in digital electronics.

One way to construct the bistable multivibrator is to use op-amps.

The op-amp bistable multivibrator circuit consists of two op-amps configured as voltage comparators, a positive feedback loop, and two resistors. The two op-amps are connected in a loop, with the output of each op-amp connected to the input of the other op-amp.

The positive feedback is what causes the circuit to switch between the two stable states. When the output of one op-amp goes high, it feeds back into the input of the other op-amp, causing its output to go low. This change in the output of the second op-amp then feeds back into the input of the first op-amp, causing its output to go low as well. This makes the circuit stable in the low state.

Conversely, if the output of one op-amp goes low, it feeds back into the input of the other op-amp, causing its output to go high. This change in the output of the second op-amp then feeds back into the input of the first op-amp, causing its output to go high as well. This makes the circuit stable in the high state.

The two resistors in the circuit are used to bias the input of each op-amp to a specific voltage level. This helps to ensure that the circuit operates reliably.

Overall, the op-amp bistable multivibrator is a simple and effective circuit for creating a stable switching behaviour. It is commonly used in digital electronics, particularly in circuits that require memory.

INTEGRATOR:

An integrator is a basic building block of analog signal processing circuits. The theory behind an integrator involves the circuit's ability to perform integration operations on input signals.

An integrator circuit can be built using an operational amplifier, a capacitor, and a resistor. The operational amplifier is configured in a feedback loop, which includes a resistor and a capacitor in series. The input signal is applied to the inverting input of the amplifier, and the output is taken from the output of the amplifier.

The capacitor in the feedback loop charges and discharges in response to changes in the input signal. The resistor limits the rate at which the capacitor charges and discharges. As a result, the output voltage of the amplifier is proportional to the integral of the input voltage over time.

The relationship between the input and output signals can be described by the transfer function of the integrator circuit. The transfer function is given by:

$H(s) = 1/(sCR)$

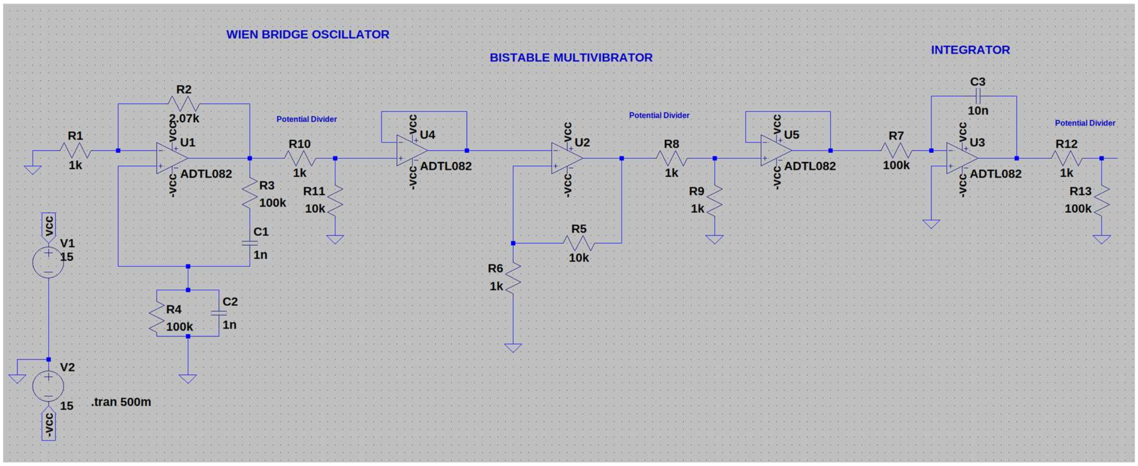
Where s is the Laplace operator, C is the value of the capacitor, and R is the value of the resistor.

The integrator circuit has a high-pass response with a cut-off frequency given by:

$f_c = 1/(2\pi RC)$

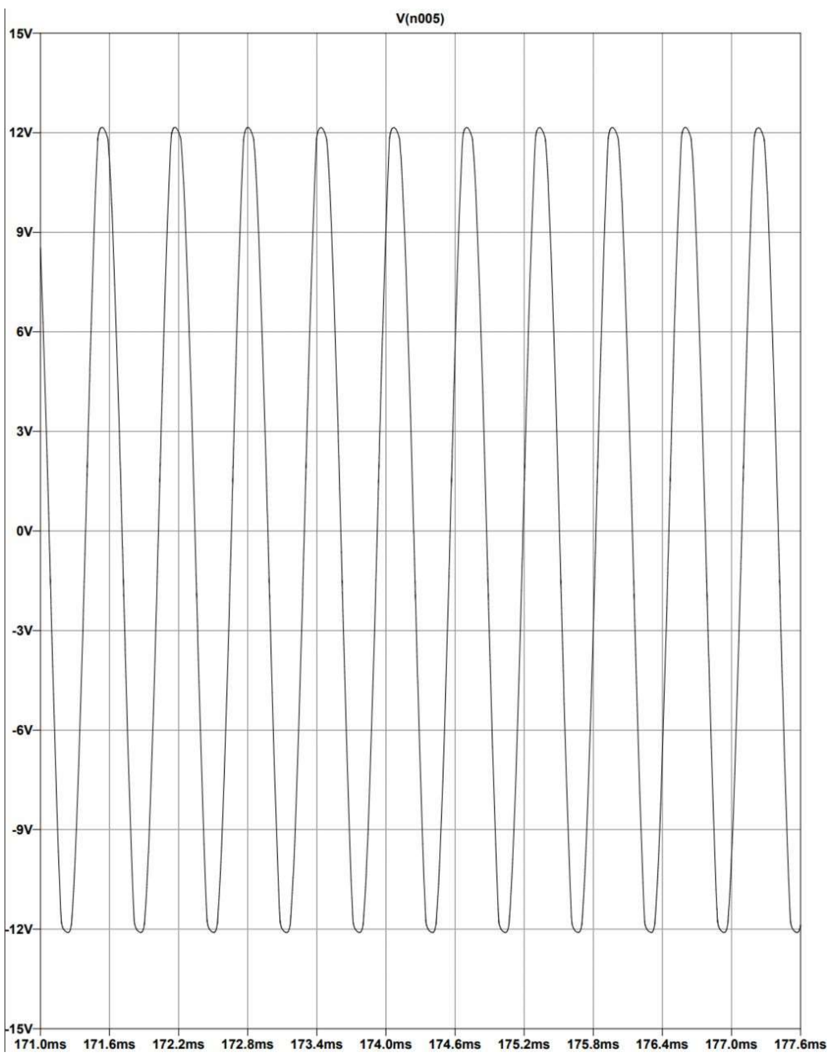
The cut-off frequency determines the frequency range for which the integrator is useful. For frequencies below the cut-off frequency, the integrator responds like a low-pass filter, while for frequencies above the cut-off frequency, the integrator acts like a differentiator.

DESIGN:

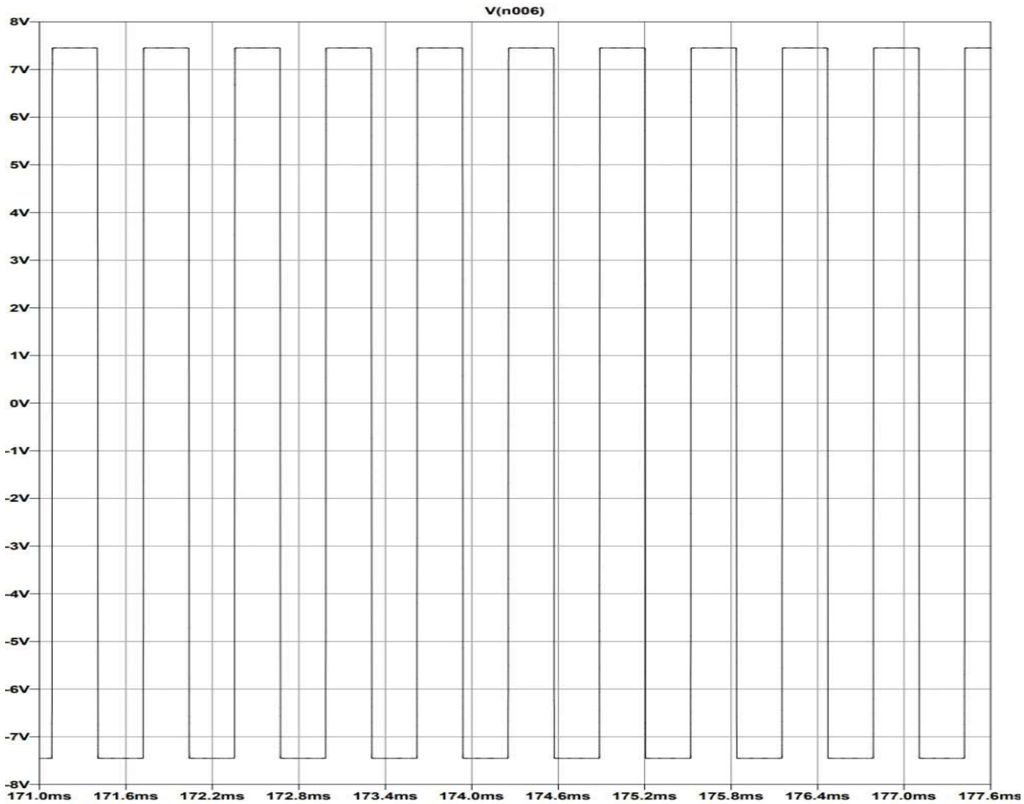


OUTPUTS:

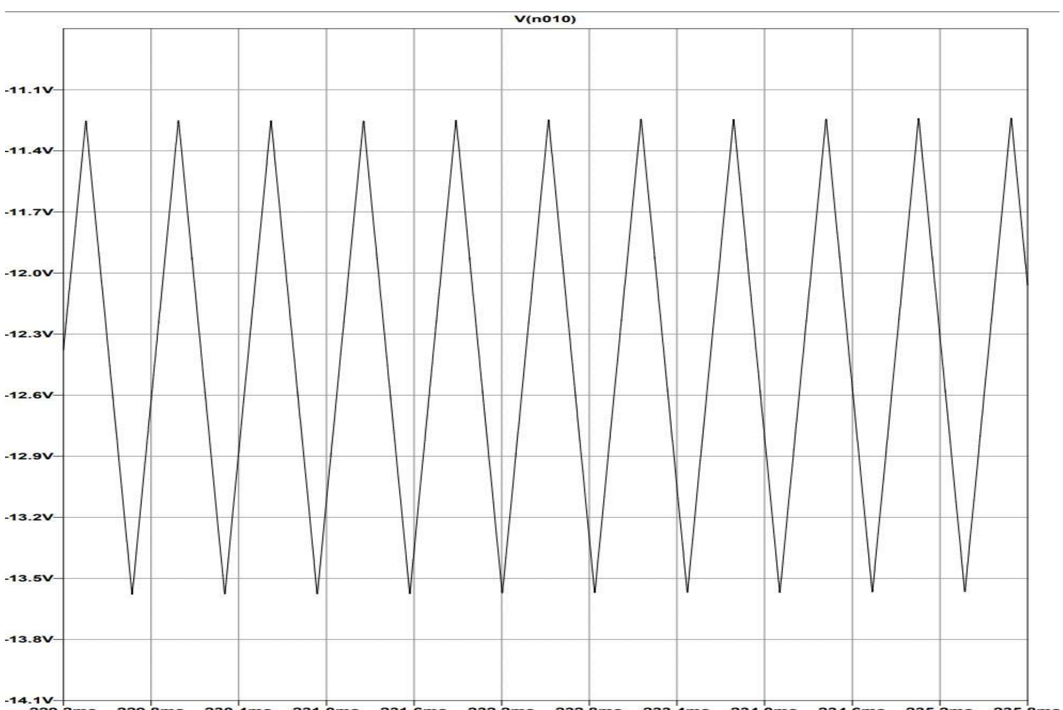
SINE WAVE:



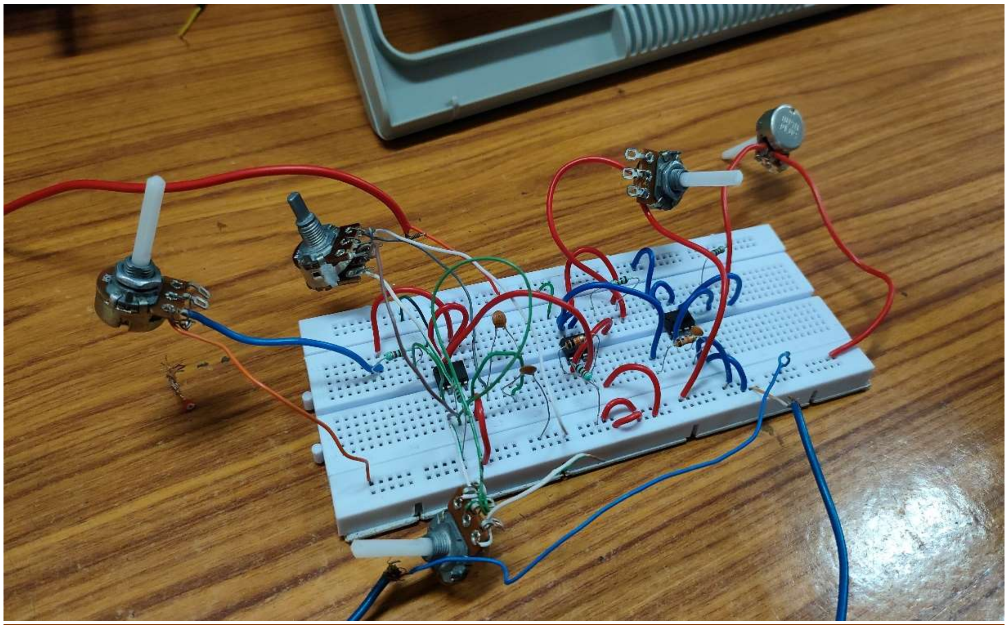
SQUARE WAVE:



TRIANGULAR WAVE:

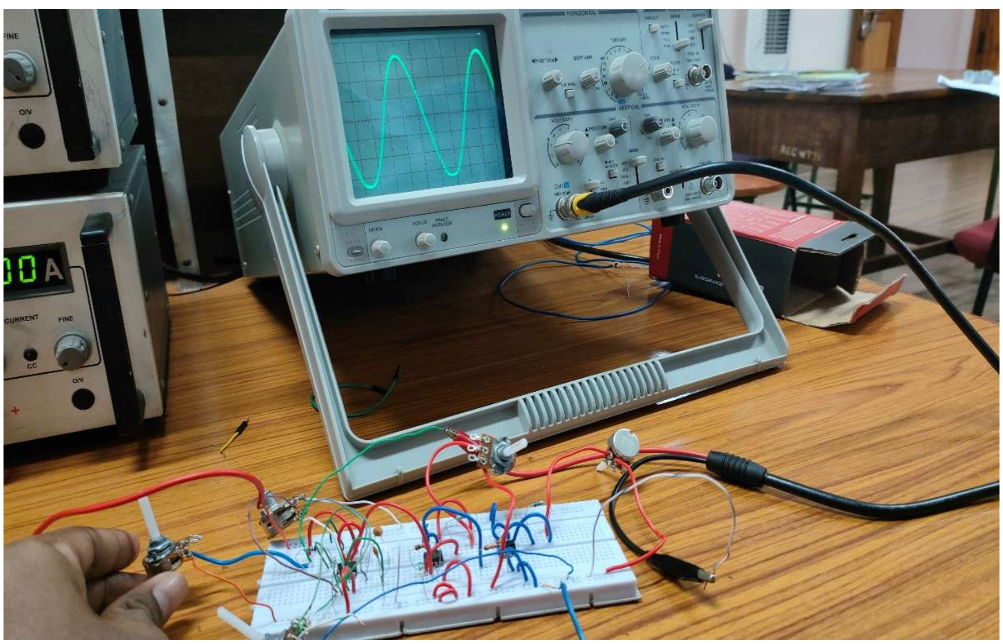


HARDWARE DESIGN:



HARDWARE OUTPUT:

SINEWAVE:



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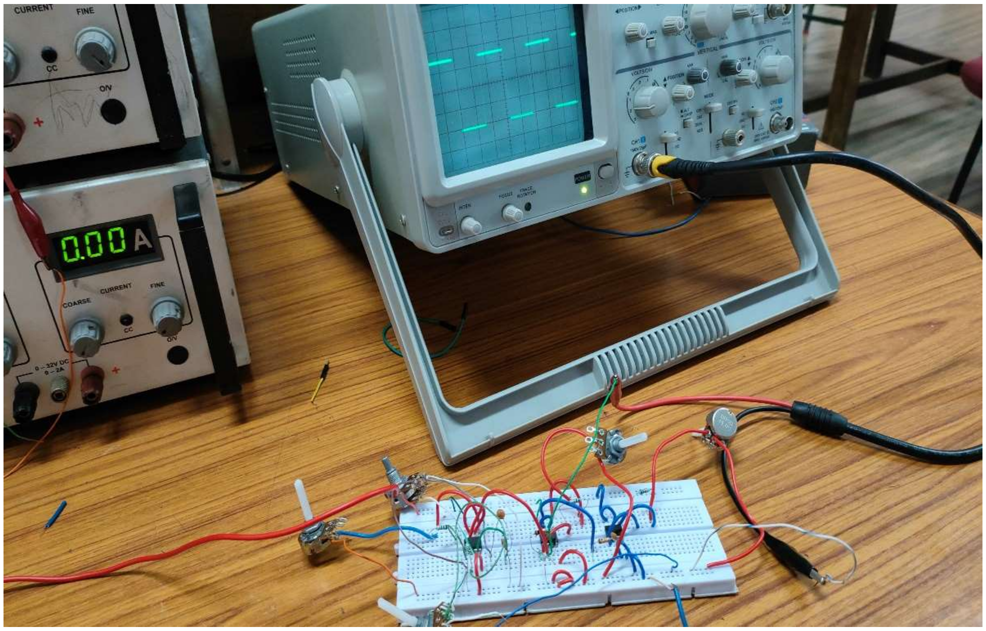

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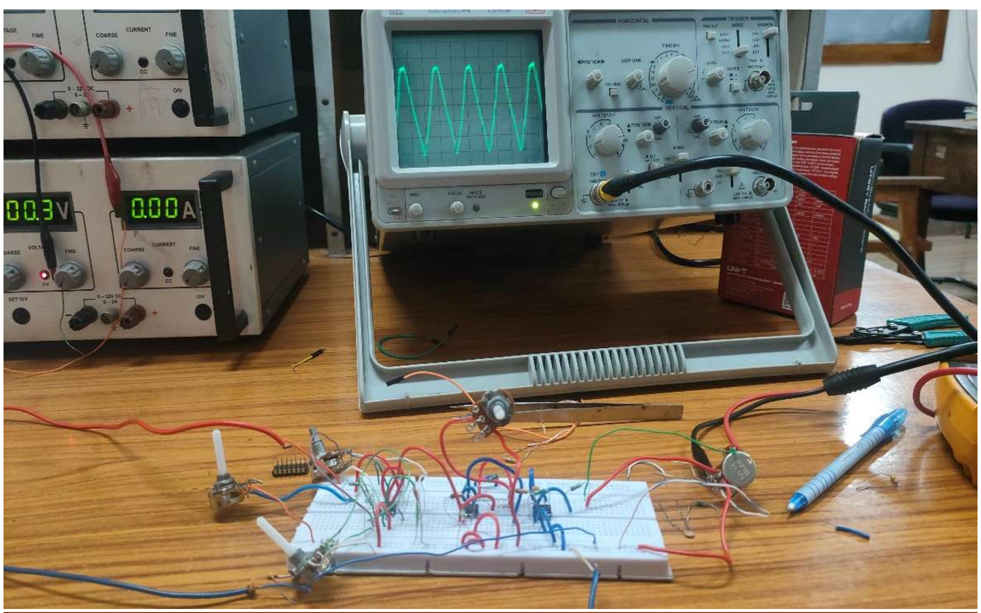

FUTURE SCOPE AND CONCLUSION:

In the future, op-amp based function generators are likely to remain an important tool for circuit designers and electronic engineers. With advancements in technology, the size and cost of op-amps are expected to decrease while their performance and versatility increase. This will enable the development of more sophisticated function generators that can generate even more complex waveforms and operate at higher frequencies.

Moreover, with the increasing demand for renewable energy sources, op-amp based function generators are likely to play a critical role in the development of renewable energy systems. Function generators can be used to control the frequency and amplitude of power signals, which is necessary for renewable energy systems to operate efficiently and safely.

In conclusion, function generators are a versatile and reliable tool for generating waveforms in electronic circuits.

REFERENCES:

EC-1 THEORY CLASS LECTURES